

AUTOSAR

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PLAN

I. Basics of Electronic Unit (ECU) in Vehicle

1. Introduction
2. ECU Software Architecture
   a. Legacy Model
   b. Challenges
   c. Solution – AUTOSAR

II. Introduction to AUTOSAR standard

1. AUTOSAR partnership
2. AUTOSAR basic principals
3. Challenges in the mobility sector
4. AUTOSAR software framework

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PLAN

III. AUTOSAR classic platform

1. Layered architecture
2. Software components
3. AUTOSAR RTE
4. AUTOSAR BSW

IV. AUTOSAR adaptive platform

1. Why do we need AUTOSAR Adaptive platform
2. AUTOSAR Classic platform VS Adaptive platform
3. AUTOSAR adaptive platform architecture
4. Functional cluster

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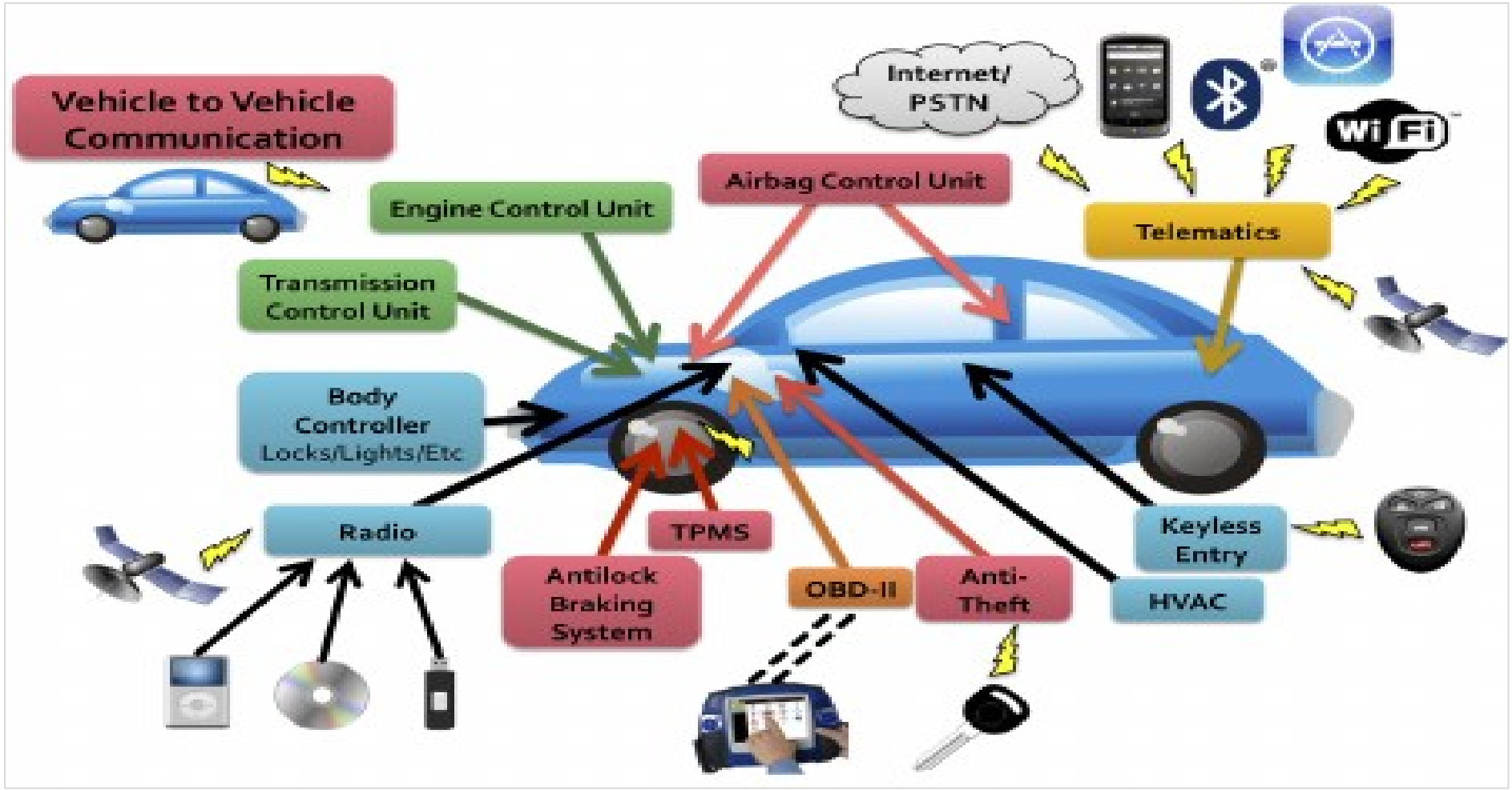
BASICS OF ELECTRONIC UNIT (ECU) IN VEHICLE

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INTRODUCTION

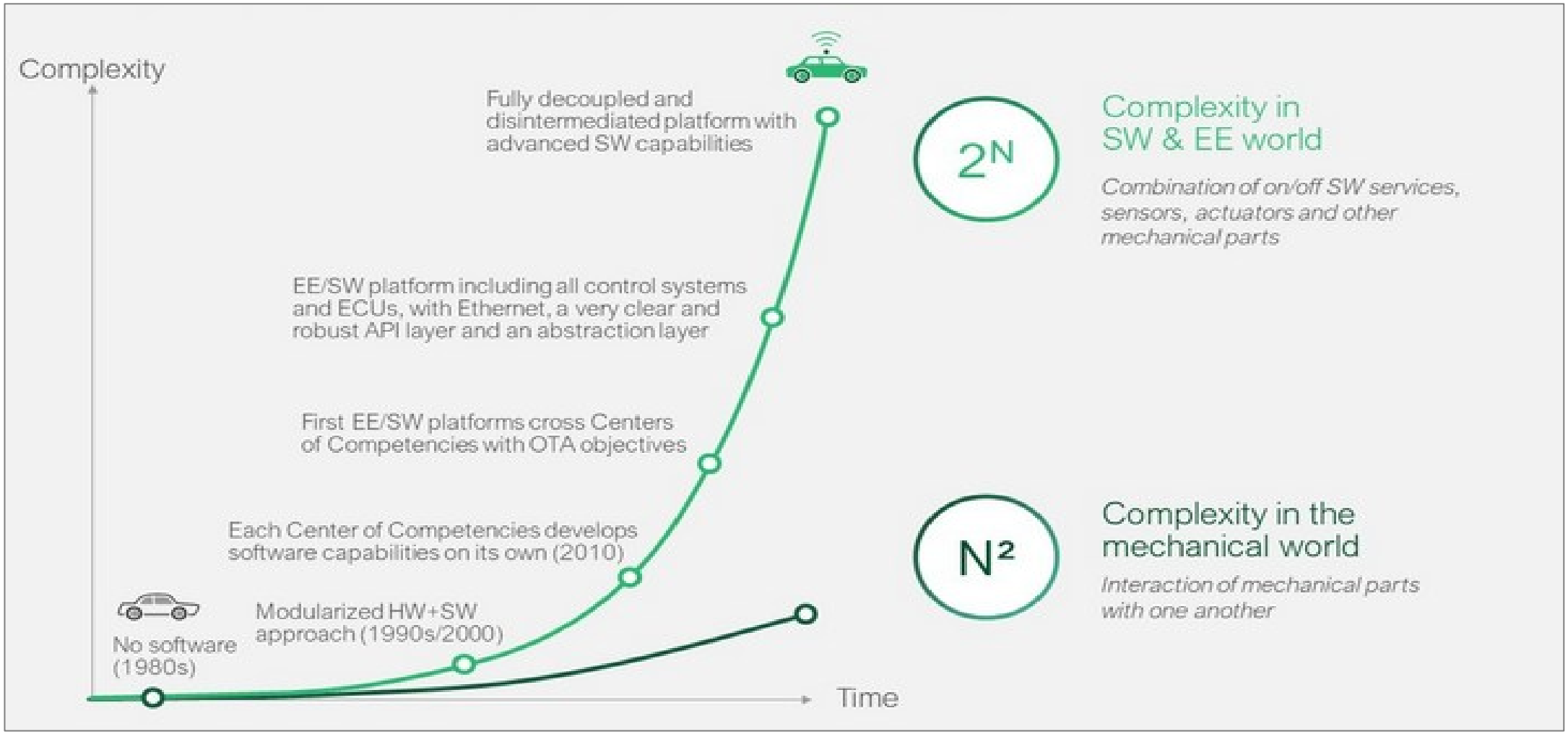
AUTOMOTIVE SYSTEMS



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INTRODUCTION

COMPLEXITY OF AUTOMOTIVE SYSTEMS

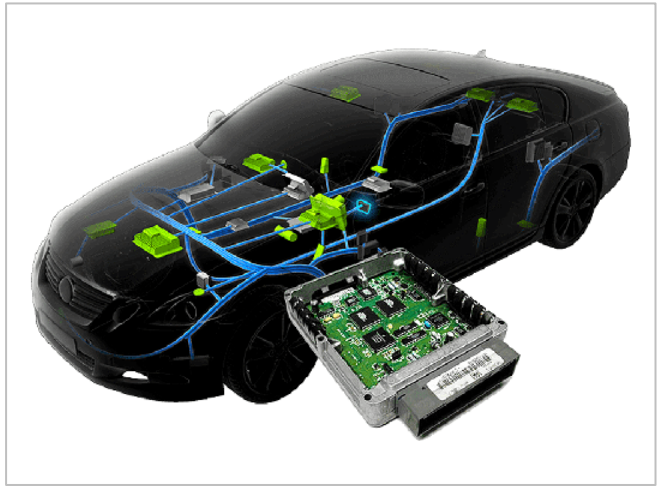


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INTRODUCTION

WHAT IS AN ECU ?

- Electronic Control Unit (ECU) is a name given to a device that controls one or more electrical systems in a vehicle.
- The ECU provides instructions for various electrical systems, instructing them on what to do and how to operate.
- Types of ECU include:
  - Electronic/engine Control module (ECM)
  - Antilock Braking Module (ABS)
  - Body Control module (BCM)

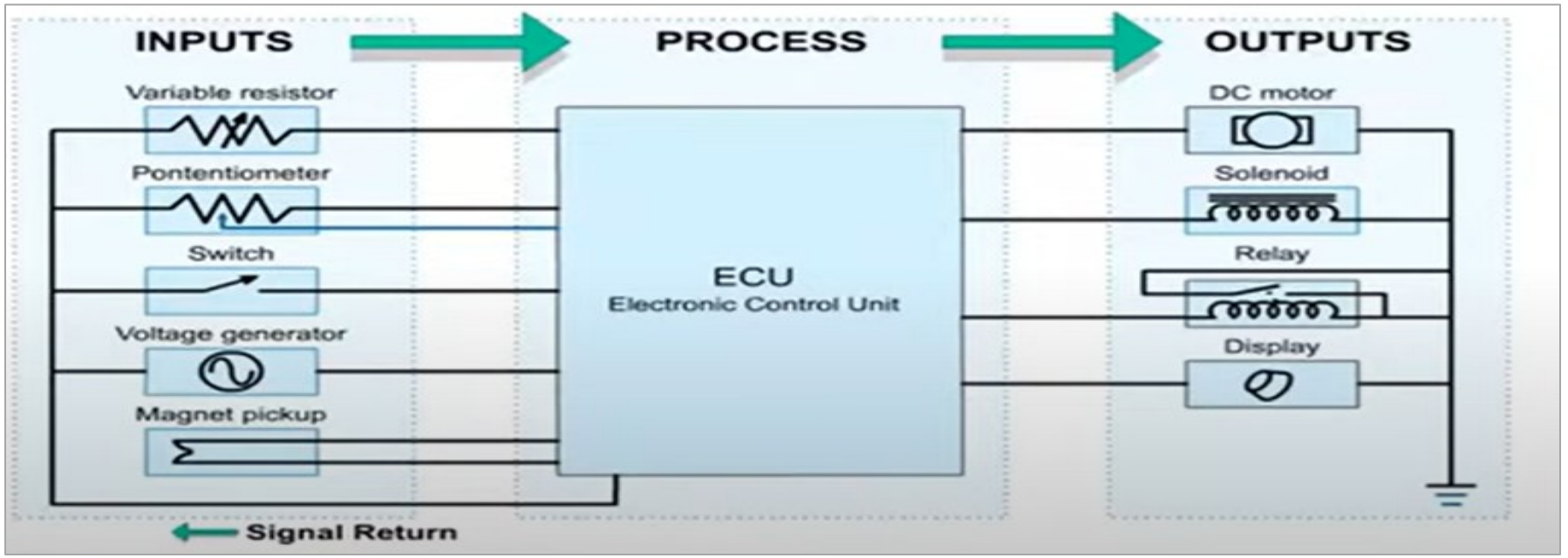


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INTRODUCTION

ECU WORKING

- Electronic Control Unit (ECU) is an electronic device has base numbers and parameters filled in its memory.
- With multiple sensors around a vehicle feeding the ECU data it can manage and control the electronic systems efficiently by giving orders to improve their output.
- Each ECU typically contains a dedicated chip that runs its own software or firmware, and requires power and data connections to operate



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ECU SOFTWARE

- ECU software is the most important component which includes the **bootloader** required to start the controller, different **configuration parameters** as well as the **decision-making logic** required to process different inputs and to provide **an accurate output**.

- The software implementing control algorithms combines the sensor values and calculates some meaningful actuator signals.

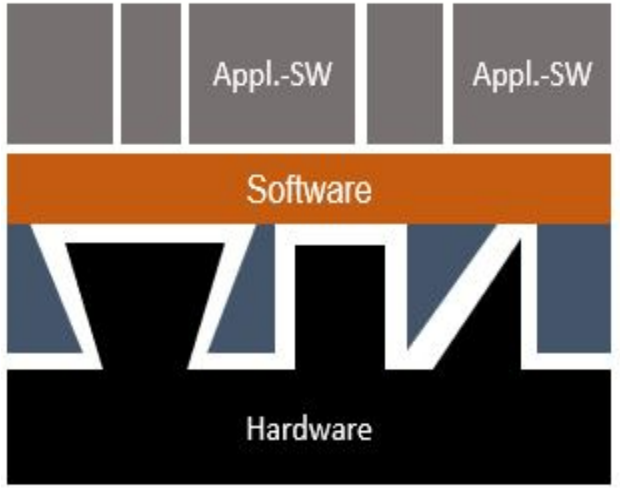
- The development itself is distributed, As a rule, a vehicle manufacturer employs several suppliers to deliver the control algorithms and the ECU. Since both the algorithms as well as the ECUs must work together, the vehicle manufacturer **must do a lot of configuration work to get the vehicle on the road.**

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LEGACY MODEL

- Each supplier has their own way of software design
- There was no commonality in the functionality supported by different ECUs
- The SW was ECU specific and hardware dependent
- Not scalable

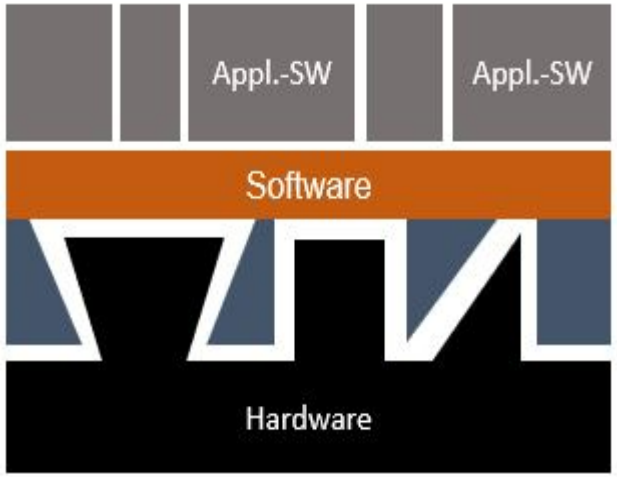


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LEGACY MODEL | CHALLENGES

- Integration of ECUs was difficult for OEMs
- Software designed for one ECU could be reused for another ECU though the supported feature is the same
- Involved more development time
- Not cost effective



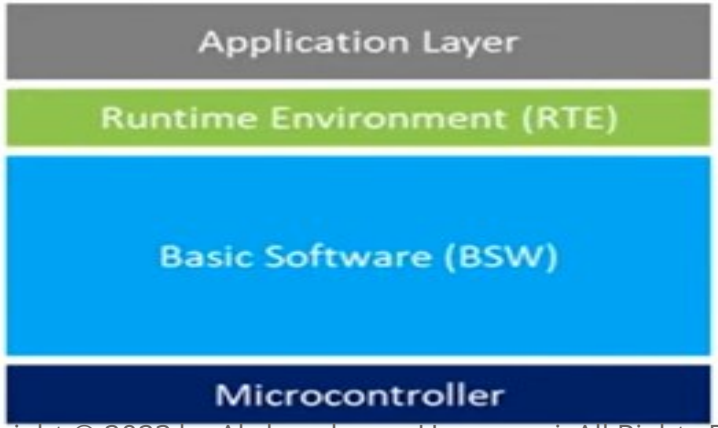
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SOLUTION - AUTOSAR

- To resolve issues in legacy model, OEMs and Tier1s produced an idea of a common standard for ECU platform software that can implement a common means of communicating with other ECUs in a network.
- Every manufacturer wants to give the best experience to customers by offering new features in their car models. A common standard gives manufacturers a possibility to focus more on application than platform software.
- OEMs share common standard for platform software, however complete in the application implementation





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SOLUTION - AUTOSAR

MAIN DRIVERS FOR AUTOSAR

- Management of E/E complexity associated with growth in functional scope
- Scalability of solutions within and across product lines
- Improved quality and reliability of E/E systems
- Flexibility for product modification, upgrade and update

AUTOSAR™

(AUTomotive Open System ARchitecture)

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AUTOSAR INTRODUCTION

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AUTOSAR MISSION



AUTOSAR is a global partnership of leading companies in the automotive and software industry to develop and establish the standardized software framework and open E/E system architecture for intelligent mobility.

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AUTOSAR VISION & PARTNERSHIP

- AUTOSAR will be the global established standard for software and methodology enabling open E/E system architectures for future intelligent mobility supporting high levels of dependability, especially safety and security.

- AUTOSAR has succeeded in bringing together main players in automotive E/E system development to form a powerful standard that is successfully used all around the world.

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MORE THAN 300 AUTOSAR PARTNERS

9 Core Partners

BOSCH, Continental, Ford, gm, PSA GROUPE, TOYOTA, VOLKSWAGEN

63 Premium Partners

APTIV, APTJ, arm, Baidu 百度, BlackBerry, Capgemini, DASSAULT SYSTEMES, Deloitte., DESAY SV automotive, dSPACE, ESOL, ETAS, Fraunhofer, Great Wall, Green Hills SOFTWARE, HCL, HELLA, HONDA, HUAWEI, HYUNDAI MOTOR GROUP, Infineon, intel, itk, Intron 英恒, CETC 普华基础软件, JTEKT, KPIT, L&T Technology Services, LG, LUXOFT, MathWorks, MITSUBISHI ELECTRIC, NEC, RENAULT, NISSAN, NXP, Panasonic, RENESAS, SCSK, SIEMENS, sodius, SUMITOMO ELECTRIC, TATA CONSULTANCY SERVICES, TATA ELXSI, TATA MOTORS, thyssenkrupp, tietoEVRY, Valeo, Tech Mahindra, VECTOR, veoneer, VOLVO, ZF

74 Development Partners

ABUP 艾拉比, Airbiquity, AirPlug, avelabs, BASELABS, basicworx, beyless, C2A, clarinox, easycore, eJad, emmtrix, ESR LABS, EVO, excelfore, Freetech, GAIO, HANECS, INCHRON, INTREPID CONTROL SYSTEMS, IP Camp, ixia, KRONO-SAFE, LAUTERBACH, MATRICKZ, SEMIDRIVE, NCES, nordsys, OPENSYNERGY, pls, PopcornSAR, PRQA, rti, RTst, RUETZ SYSTEM SOLUTIONS, SAFERIDE, HINGE MYNDS, softing, SYSGO, TECHNO-WELLE, tracetronic, TTTech, VALIDAS, VAULT MICRO, VEROLT, UNIKIE, 博云 BeCloud, Acsia

1 Strategic Partner

DENSO
Crafting the Core

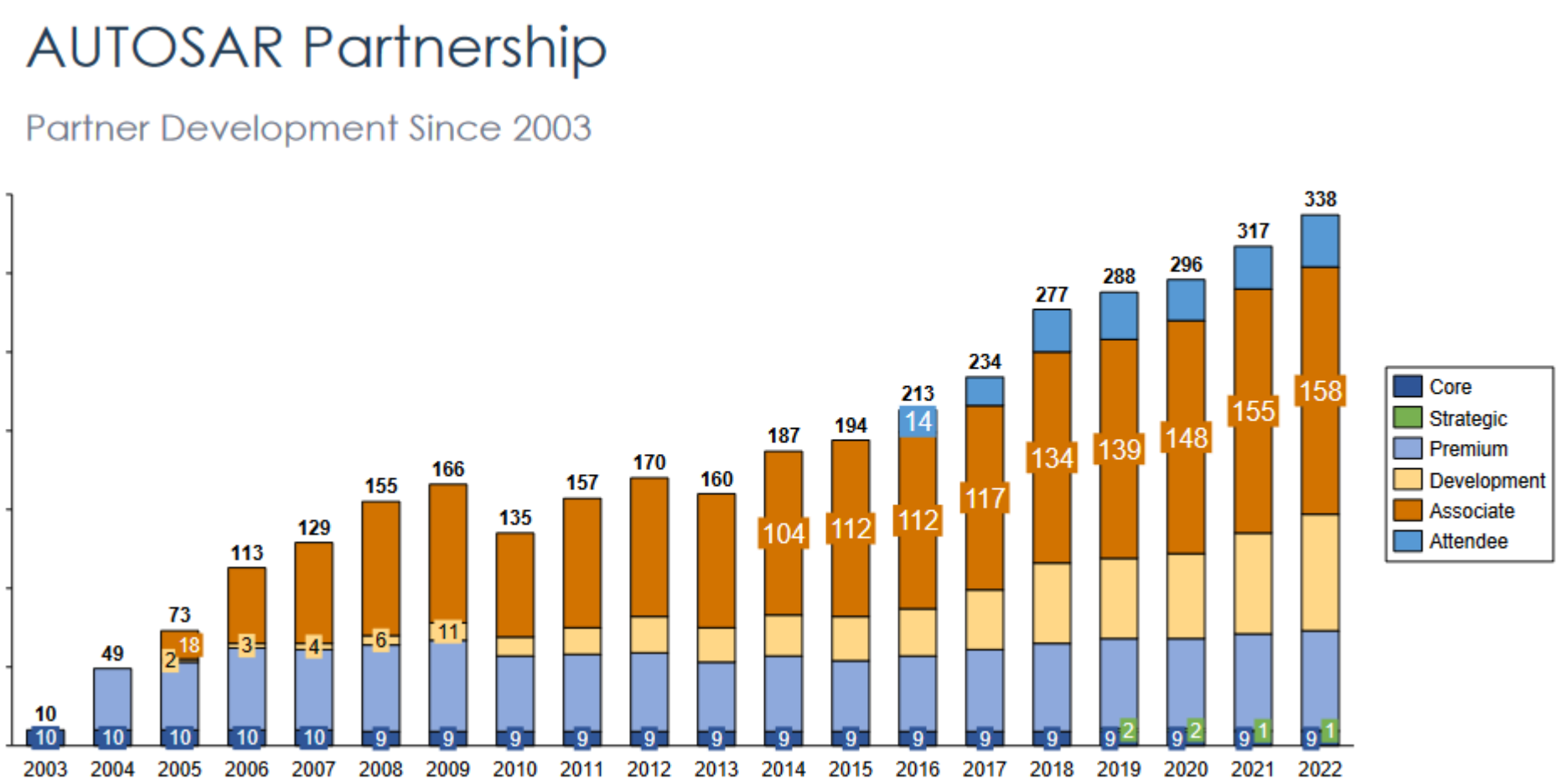
+ 158
Associate Partners

+ 33
Attendees

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HISTORY AND CURRENT STATE



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GEOGRAPHICAL DISTRIBUTION OF PARTNERS

AUTOSAR Partnership

Global Distribution of AUTOSAR Partners



38 Partners in North America
- 2 Core Partner
- 6 Premium Partner
- 6 Development Partner
- 23 Associate Partner
- 1 Attendee

153 Partners in Europe
- 6 Core Partner
- 29 Premium Partner
- 38 Development Partner
- 55 Associate Partner
- 25 Attendee

144 Partners in Asia
- 1 Core Partner
- 1 Strategic Partner
- 28 Premium Partner
- 28 Development Partner
- 80 Associate Partner
- 6 Attendee

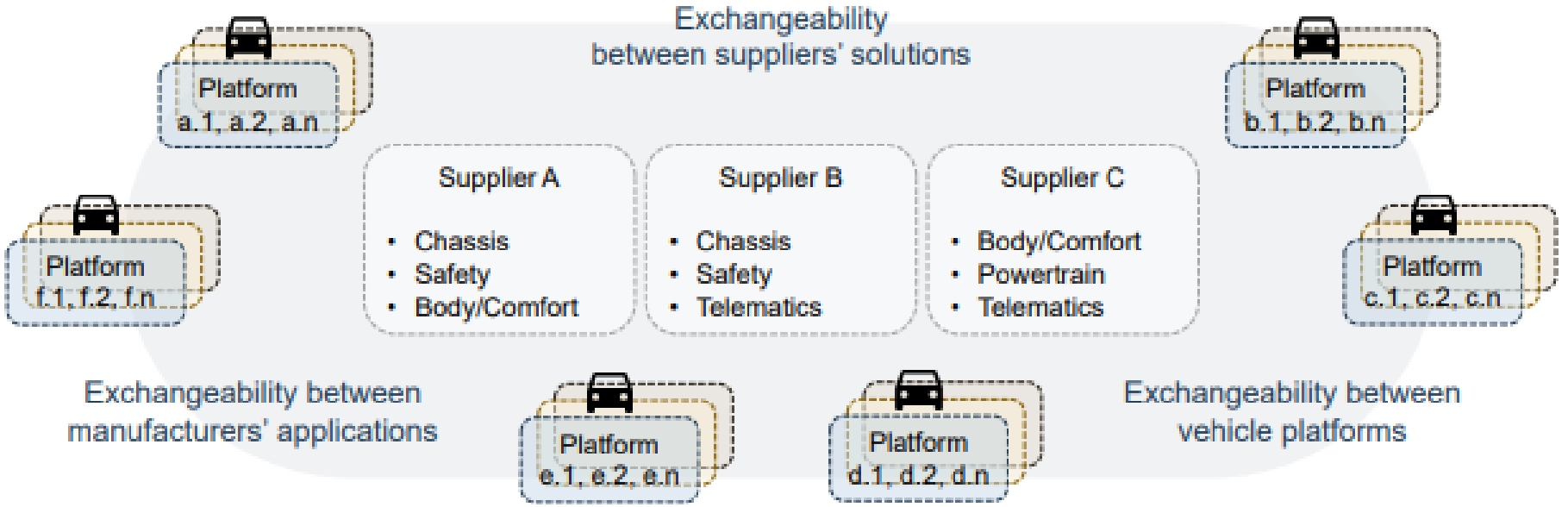
3 Partners in Africa
- 2 Development Partner
- 1 Attendee

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AUTOSAR BASIC PRINCIPALS

Benefits of a Software Framework



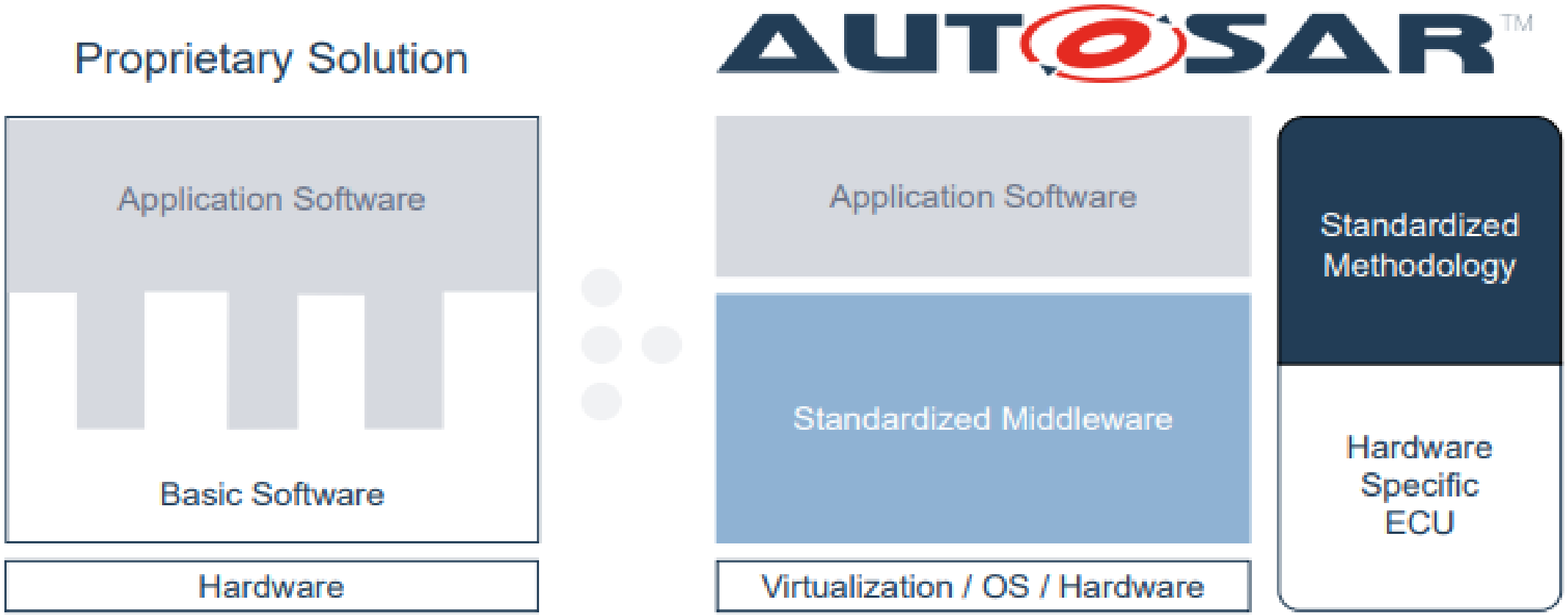
The AUTOSAR Software Framework promotes software module **reuse and exchangeability.**

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AUTOSAR BASIC PRINCIPALS

Proprietary vs. AUTOSAR Middleware Approach

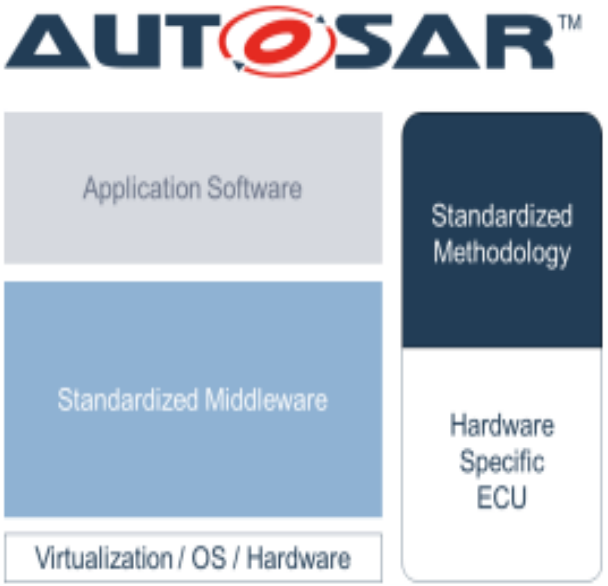


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Benefits of the AUTOSAR Middleware Approach

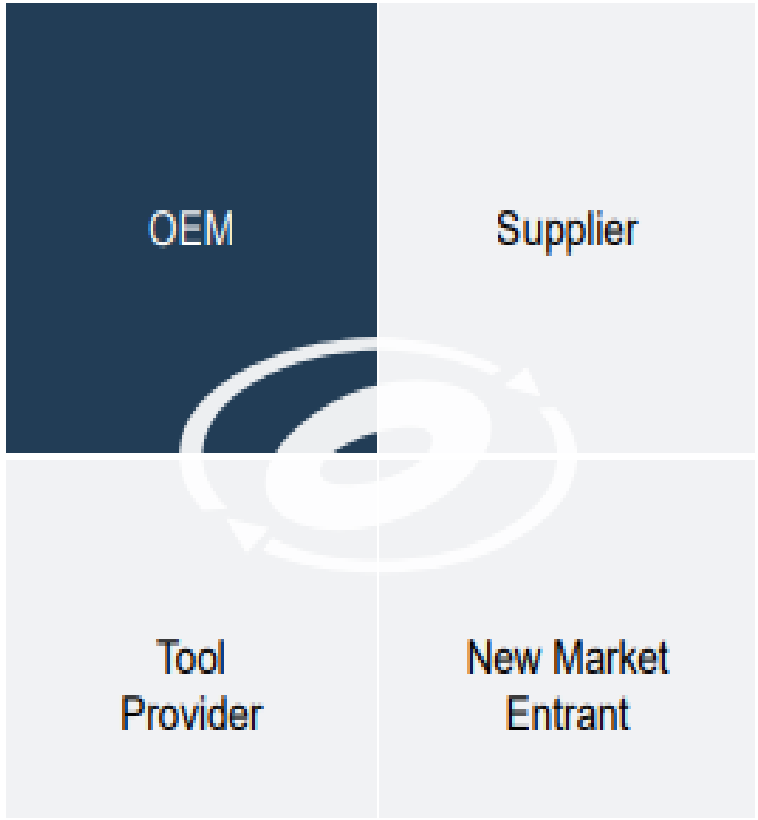
AUTOSAR paves the way for innovative electronic systems with improved performance, safety and security.

- ❑ Hardware and software widely independent of each other
- ❑ Decouplable development (by abstraction) through horizontal layers; therefore, reduced development time and costs
- ❑ Enhanced quality and efficiency through software reuse



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Benefits of Exploiting the Standard (1)



- ❑ Establish distributed development among suppliers
- ❑ Compete on innovative functions with increased design flexibility
- ❑ Simplify software and system integration
- ❑ Reduce overall software development costs

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Benefits of Exploiting the Standard (2)



- ❑ Enhance efficient variant handling
- ❑ Reuse software modules across OEMs
- ❑ Increase efficiency of application development
- ❑ Invent new business models

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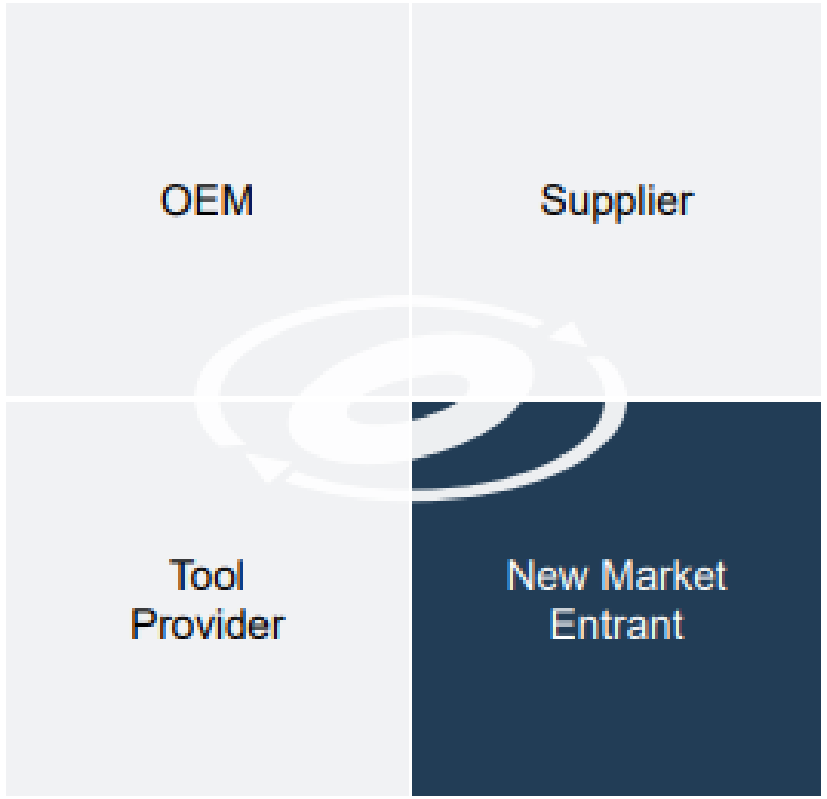
Benefits of Exploiting the Standard (3)



- Have an interface with development processes
- Embed tools into an overall tool environment

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Benefits of Exploiting the Standard (4)



- ❑ Enable new business models through standardized interfaces
- ❑ Understand easily how automotive software is developed

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CHALLENGES IN THE MOBILITY SECTOR

Selected Main Drivers for Standardization



Highly Automated Driving with Dependability

- Reliability
- Availability
- Maintainability
- Safety
- Security



V2X, Internet of Things, Cloud-Based Services

- Security
- QoS
- Over the Air (OTA) Update/Upgrade

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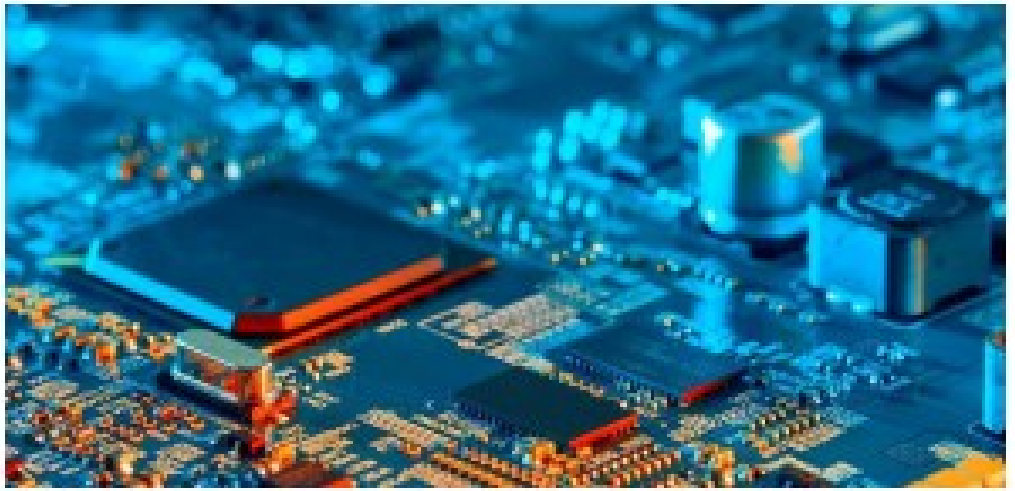
CHALLENGES IN THE MOBILITY SECTOR

Selected Main Drivers for Standardization



Increasing Data Rates and Volume

- Automotive Ethernet
- 5G



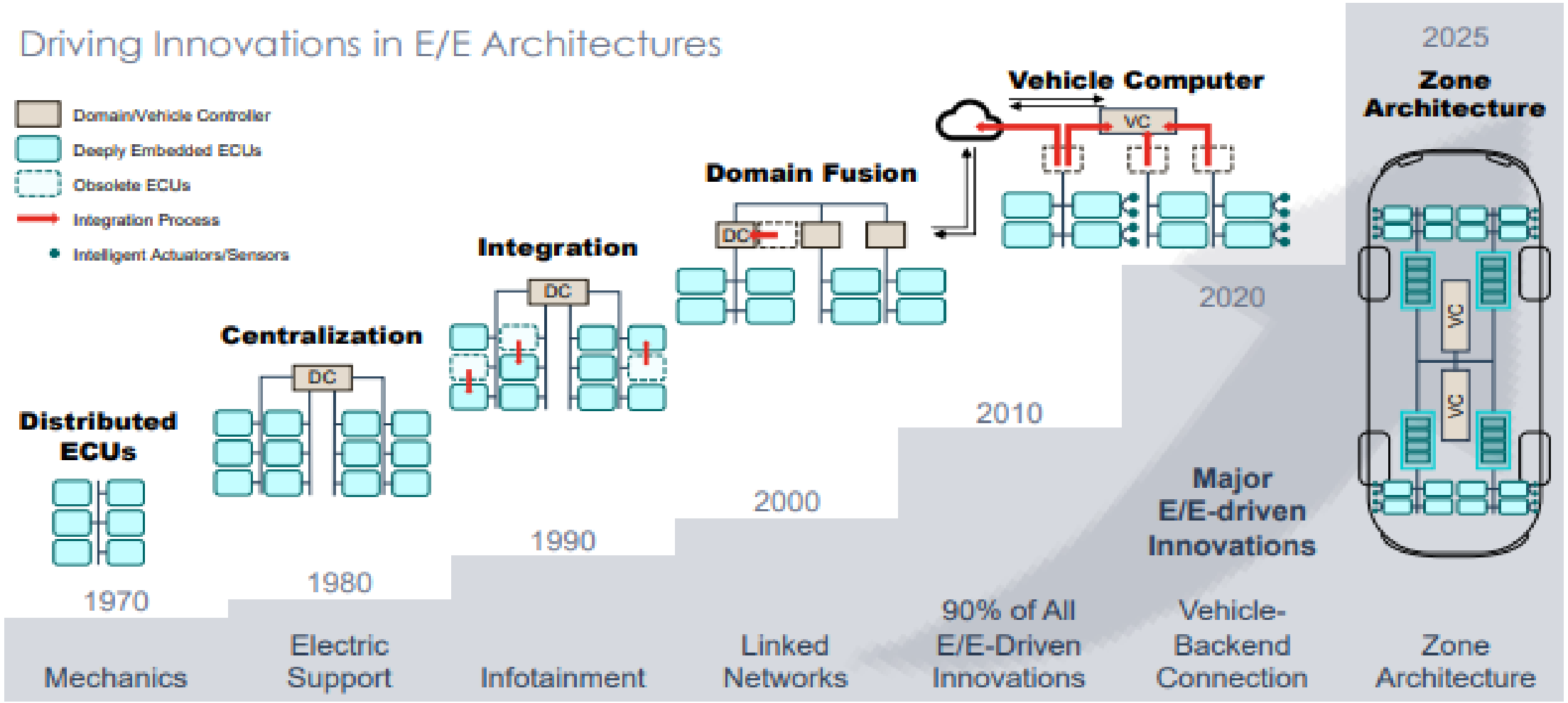
New Automotive Processor Technologies

- Centralized multi-core processors

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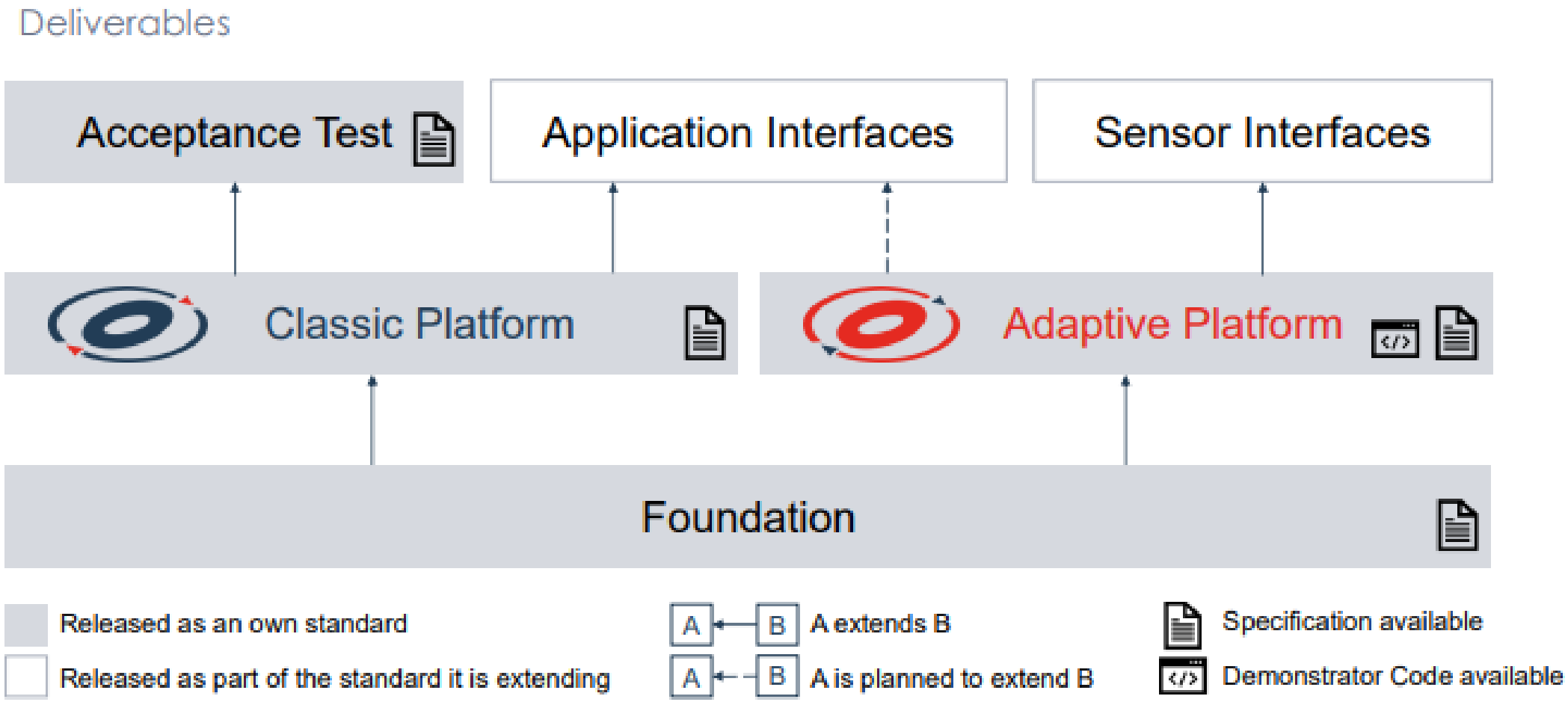
CHALLENGES IN THE MOBILITY SECTOR



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AUTOSAR SOFTWARE FRAMEWORK



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Common features

The Foundation assures compatibility of the different AUTOSAR standards and therefore contains all common artifacts such as:

- **Main requirements:** common requirements for classic and adaptive AUTOSAR platforms
- **Meta model:** provides means to describe the AUTOSAR architecture with all its interfaces
- **Protocols:** provides standardized specification of communication protocols such as SOME/IP
- **Debugging specification (DLT)**

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The AUTOSAR Platforms

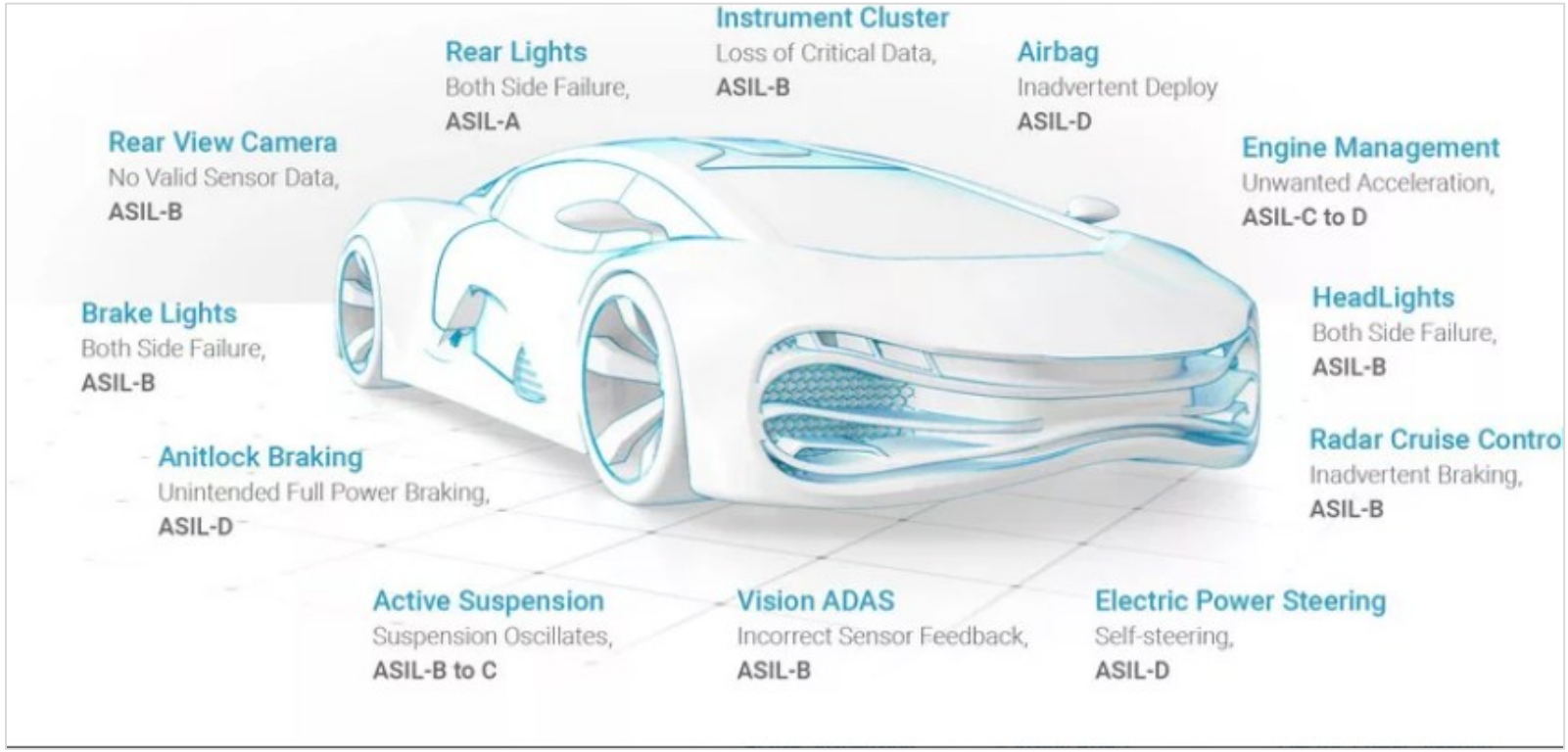
	Classic Platform	Adaptive Platform	Collaboration E.g. Infotainment
			Microsoft Windows, Android, Linux, Automotive Grade Linux, GENIVI, Robot Operating System (ROS)
Real Time Requirements	**High**, in the range of micro-seconds	**Mid**, in the range of milli-seconds	**Low**, in the range of seconds
Safety Criticality	**High**, up to ASIL-D	**High**, at least ASIL-B	**Low**, QM
Computing Power	**Low**, ~ 1000 DMIPs	**High**, > 20.000 DMIPs	**High**, ~ 10.000 DMIPs

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INTRODUCTION

ASIL INTRODUCTION

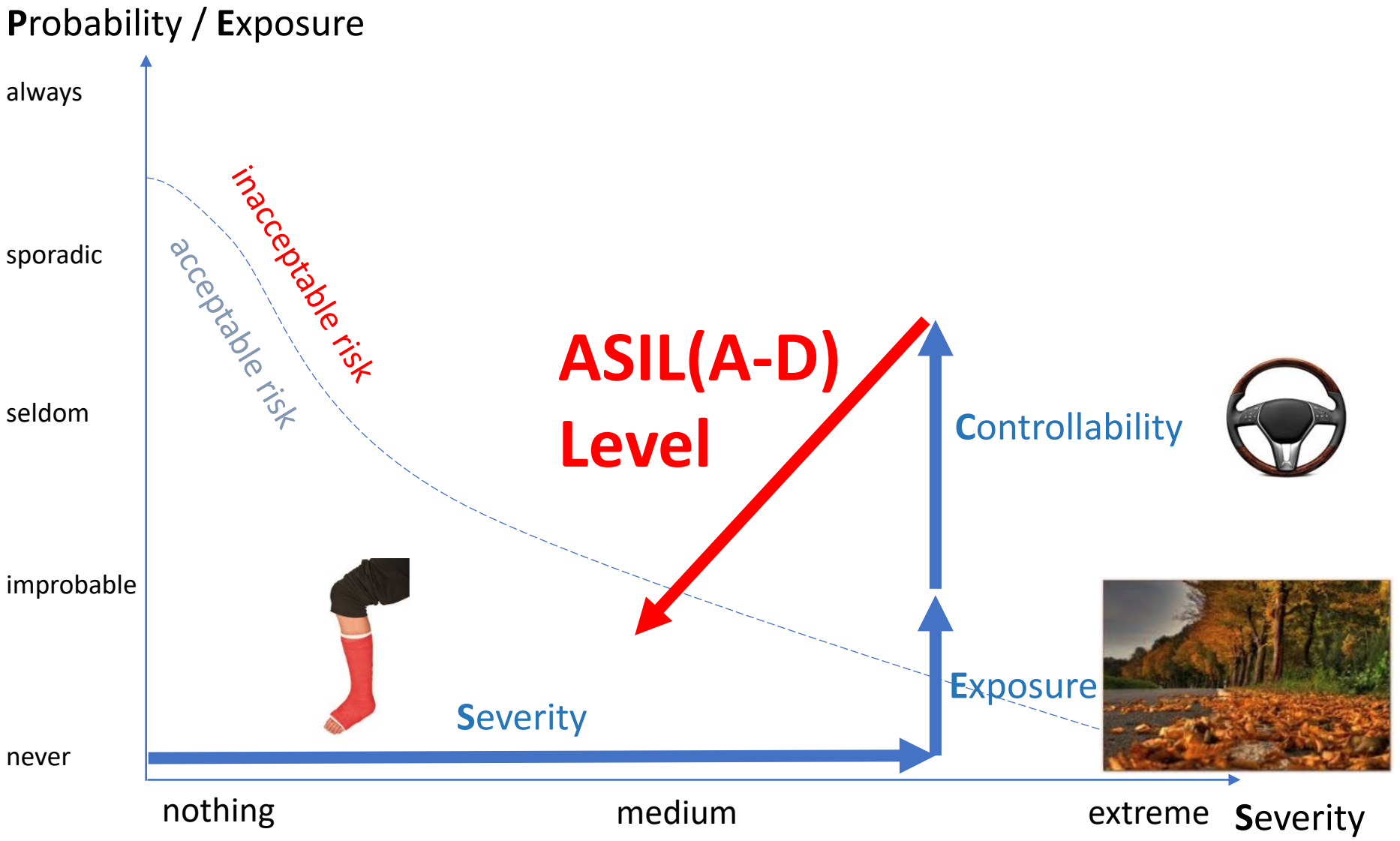
- ASIL refers to **Automotive Safety Integrity Level**. It is a risk classification system defined by the ISO 26262 standard for the functional safety of road vehicles.
- There are four ASILs identified by ISO 26262—A, B, C, and D. ASIL A represents the lowest degree and ASIL D represents the highest degree of automotive hazard.



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ASIL LEVELS



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ASIL TERMS AND DEFINITION

- **Severity:** estimate of the extent of harm (physical injury or damage to the health of persons) to one or more individuals that can occur in a potentially hazardous situation

- **Exposure:** state of being in an operational situation that can be hazardous if coincident with the failure mode under analysis

- **Controllability:** ability to avoid a specified harm or damage through the timely reactions of the persons involved, possibly with support from external measures. Persons involved can include the driver, passengers or persons in the vicinity of the vehicle's exterior.

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AUTOSAR CLASSIC PLATFORM

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AUTOSAR CLASSIC PLATFORM

LAYERED ARCHITECTURE

- The AUTOSAR Architecture distinguishes on the highest abstraction level between three software layers:
  - Application
  - Runtime Environment
  - Basic Software

  which run on a Microcontroller.



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AUTOSAR CLASSIC PLATFORM

LAYERED ARCHITECTURE

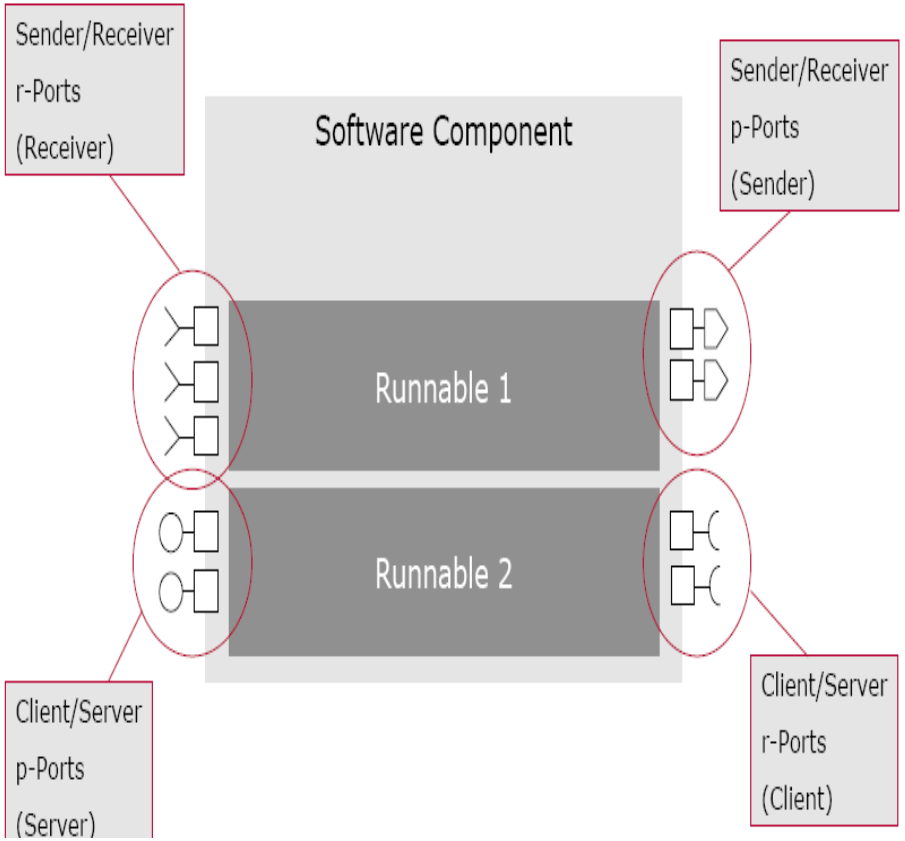
- **Application:** Defined Software Components (SWCs) with clearly specified interfaces to the Basic Software (BSW)
- **Basic Software:** The modules of the BSW provide fundamental standard services such as bus communications, memory management, IO access,…..
- **The runtime environment or RTE:** controls the connections between SWCs and from SWCs to the BSW. It provides the mechanisms required to make the VFB abstraction work at runtime. The RTE is, therefore, in the simplest case, an implementation of the VFB.
- The Virtual Functional Bus (VFB) specified by AUTOSAR represents the conceptual foundation for communication between SWCs and for the use of BSW services. The SWCs are independent of the ECU hardware.

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AUTOSAR CLASSIC PLATFORM

AUTOSAR SW-C (SOFTWARE COMPONENTS)

- Application is divided into SW-Cs
- Software Components consist of :
  - **Ports:** Interface to other SW-Cs
  - **Runnable Entities (or Runnables):** Procedures which contain the actual implementation triggered cyclically or on event (e.g. data reception)

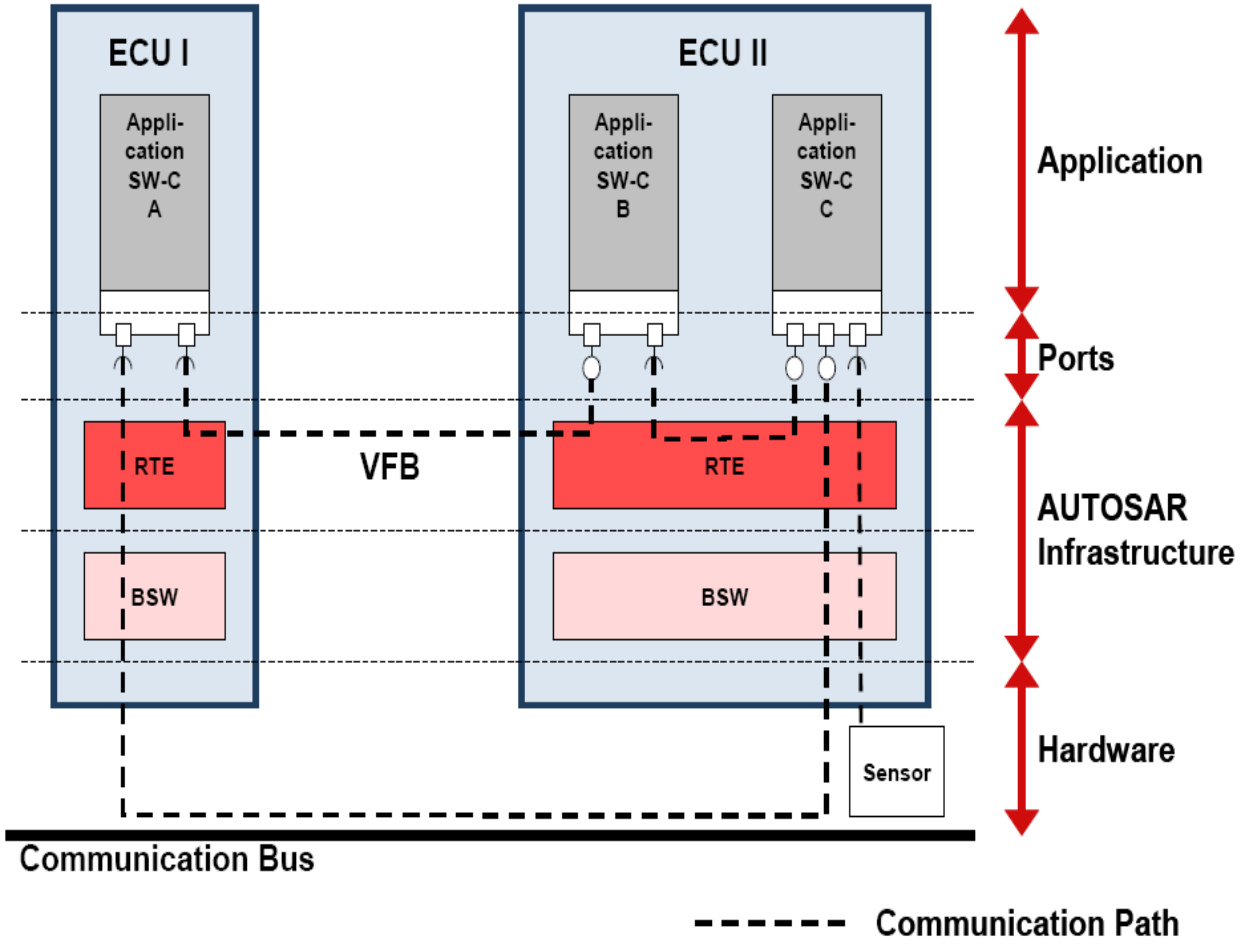


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AUTOSAR CLASSIC PLATFORM

AUTOSAR RTE (RUNTIME ENVIRONEMENT)

- Implementation of Virtual Functional Bus
- Interface between SW-Cs and Basic Software
- All calls to basic software pass through the RTE
- Communication method : Send/Receive signals, Client/Server functionality
- Triggering of runnables : Cyclically or On event



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- A Port Interface characterizes the information provided or required by a port of a Software Component.

- A Port Interface is either a :
  - **Client-Server Interface:** in case client-server communication is chosen.
  - **Sender-Receiver Interface:** in case sender-receiver communication is used.

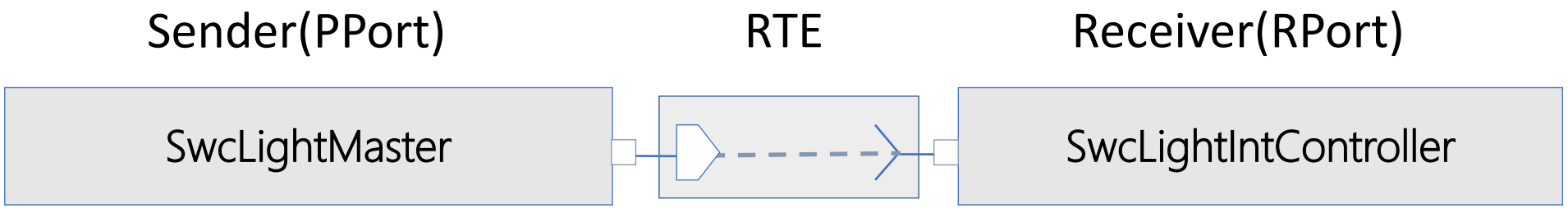
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- A sender-receiver (S/R) interface is a special kind of port-interface used for the case of sender-receiver communication.
  - **The sender-receiver interface** defines the data-elements which are sent by a sending component (which has a p-port providing the sender-receiver interface) or received by a receiving component (which has an r-port requiring the sender-receiver interface).
  - **Sender-Receiver interfaces** are used for data exchange between software components.

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AUTOSAR CLASSIC PLATFORM

SENDER-RECEIVER INTERFACE | EXAMPLE




```
PPort ctrFn
{
 DataElem ctrFn1SrRdlDRD    [PrimitiveType UInt8]
 DataElem ctrFn1SrRdlPSD [PrimitiveType UInt8]
 DataElem ctrFn1SrIntl [PrimitiveType UInt8]
 DataElem ctrFn2SrRdlDRD    [PrimitiveType UInt8]
 DataElem ctrFn2SrRdlPSD [PrimitiveType UInt8]
 DataElem ctrFn2SrIntl [PrimitiveType UInt8]
}
```

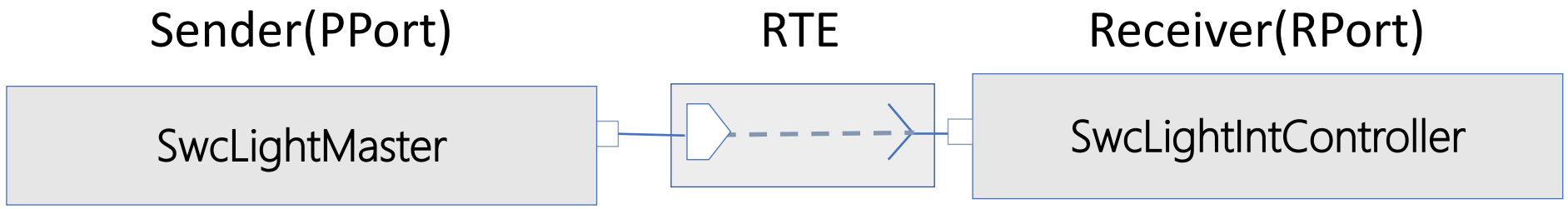
```
RPort ctrFn
{
 DataElem ctrFn1SrRdlDRD    [PrimitiveType UInt8]
 DataElem ctrFn1SrRdlPSD [PrimitiveType UInt8]
 DataElem ctrFn1SrIntl [PrimitiveType UInt8]
 DataElem ctrFn2SrRdlDRD    [PrimitiveType UInt8]
 DataElem ctrFn2SrRdlPSD [PrimitiveType UInt8]
 DataElem ctrFn2SrIntl [PrimitiveType UInt8]
}
```

```
Composition SwcLightComposition{
            SwcLightMaster
            SwcLightIntController
Connect{
            SwcLightIntController.ctrFn with SwcLightMaster.ctrFn
}
}
```

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AUTOSAR CLASSIC PLATFORM

SENDER-RECEIVER INTERFACE | SW IMPLEMENTATION EXAMPLE



OutputSignals.h

- **Provide information**

```
void ctrFn_ctrFn1SrRdlDRD
(UInt8 valueToWrite) const
{
  Rte_Write_ctrFn_ctrFn1SrRdlDRD (valueToWrite);
}
```

LicSwcInterface.cpp

- **Receive information from LightMaster**

```
void LicSwcInterface::ReadFunctionsInput
(bool a_bReadOnlyChanged, IInputFunctionsHandler&
a_handler)
{
TUInt8 value;
Rte_Read_ctrFn_ctrFn1SrRdlDRD(&value);
}
```

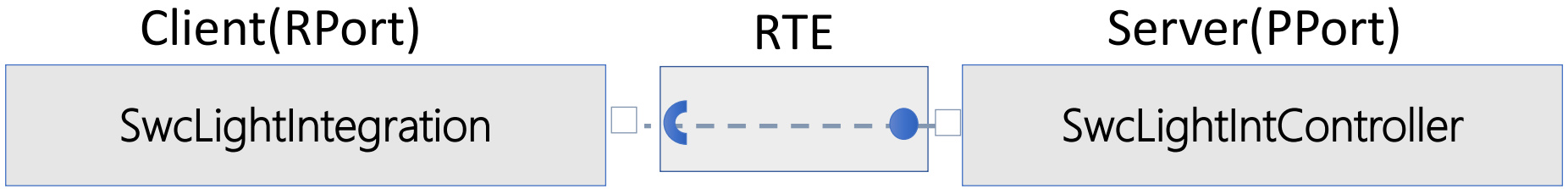
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- The client-server interface is a special kind of port-interface used for the case of Client-Server Communication.
- The client-server interface defines the operations that are provided (implemented) by the server and that can be used by the client.
- Client-server interfaces allow a client to call an operation at a server, which in turn provides the result to the client.
- The argument data prototypes of the operation of an interface can be the client input to the server (IN) or the server result (OUT) or both (INOUT).

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AUTOSAR CLASSIC PLATFORM

CLIENT-SERVER INTERFACE - EXAMPLE




```
RPort isOutputChannelConfigured
{
 Operation isOutputChannelConfigured
 {
  Argument lightDomain [PrimitiveType
Lic_Light_Domain][IN]
  Argument ocId [PrimitiveType UInt8] [IN]
  Argument result [PrimitiveType Lic_Result] [OUT]
 }
}
```

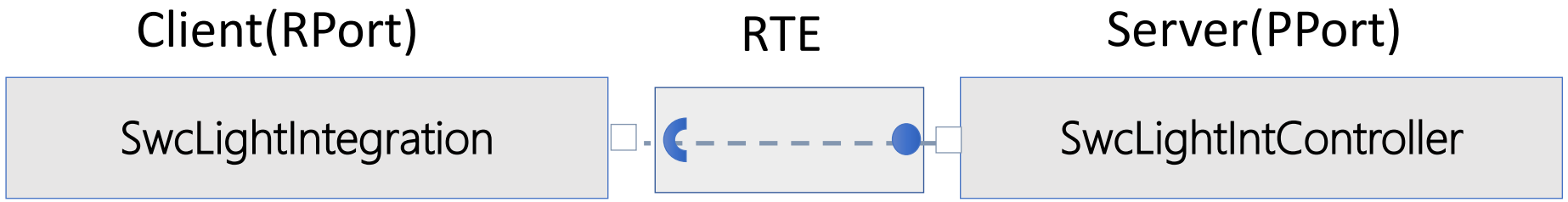
```
PPort isOutputChannelConfigured
{
Operation isOutputChannelConfigured
{
Argument lightDomain [PrimitiveType Lic_Light_Domain] [IN]
Argument ocId [PrimitiveType UInt8] [IN]
Argument result [PrimitiveType Lic_Result] [OUT]
}
}
```

```
Composition SwcLightComposition{
            SwcLightIntegration
            SwcLightIntController
Connect{
            SwcLightIntegration.isOutputChannelConfigured with SwcLightIntController.isOutputChannelConfigured
}
}
```

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AUTOSAR CLASSIC PLATFORM

CLIENT-SERVER INTERFACE – SW IMPLEMETATION



LciDiagJobs.h

- **Invoke method**

```
inline static uint8 getOutputConfigurationLic
( uint8 outputChannelId)
{
uint8 result = LIC_RESULT_FALSE;
Rte_Call_isOutputChannelConfigured_isOutputChannel
Configured(&result, LIC_INTERIOR_LIGHT_DOMAIN,
outputChannelId);

}
```

isOutputChannelConfiguredRunnable.c

- **Receive request and send response**

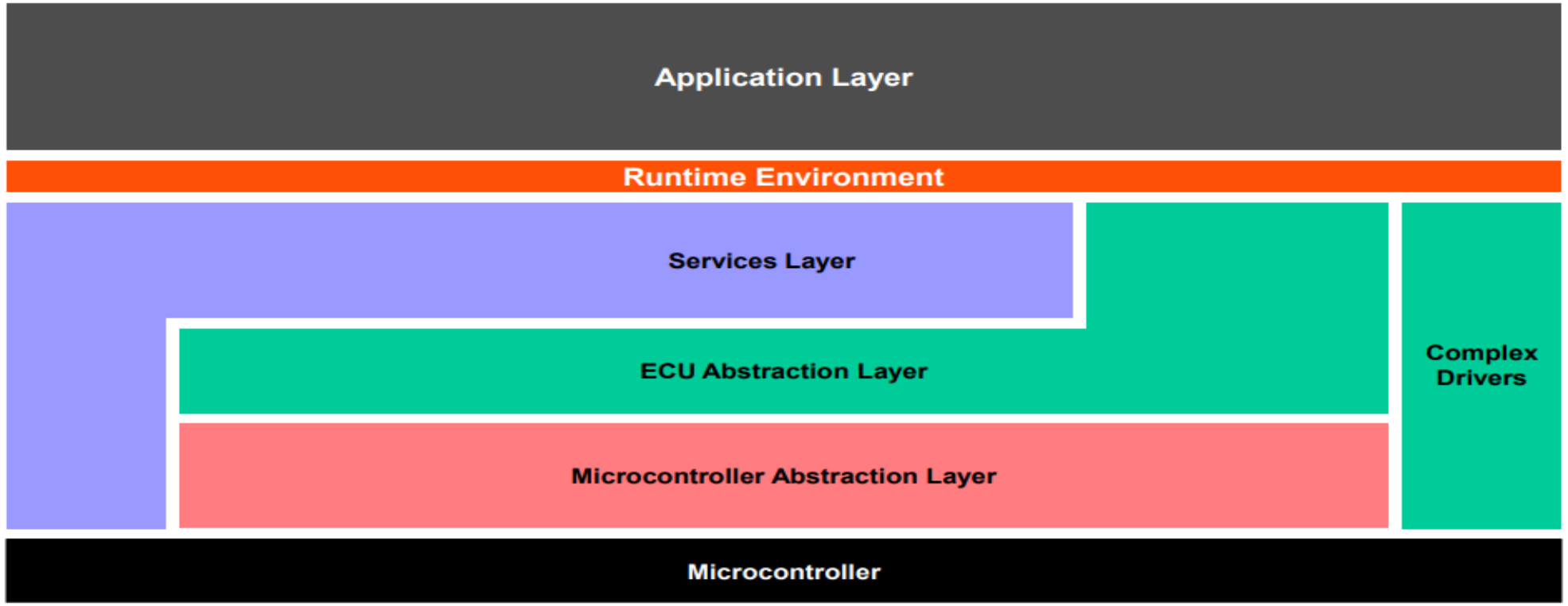
```
FUNC(void, SWCLIGHTINTENSITYCONTROLLER_CODE)
RTERunnable_SwcLightIntController_isOutputChannelConfigured
(Lic_Result* result, Lic_Light_Domain lightDomain, UInt8 ocId)
{
   if(result != NULL)
   {
      *result = Lic_isOutputChannelConfigured(lightDomain, ocId);
   }
}
```

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AUTOSAR CLASSIC PLATFORM

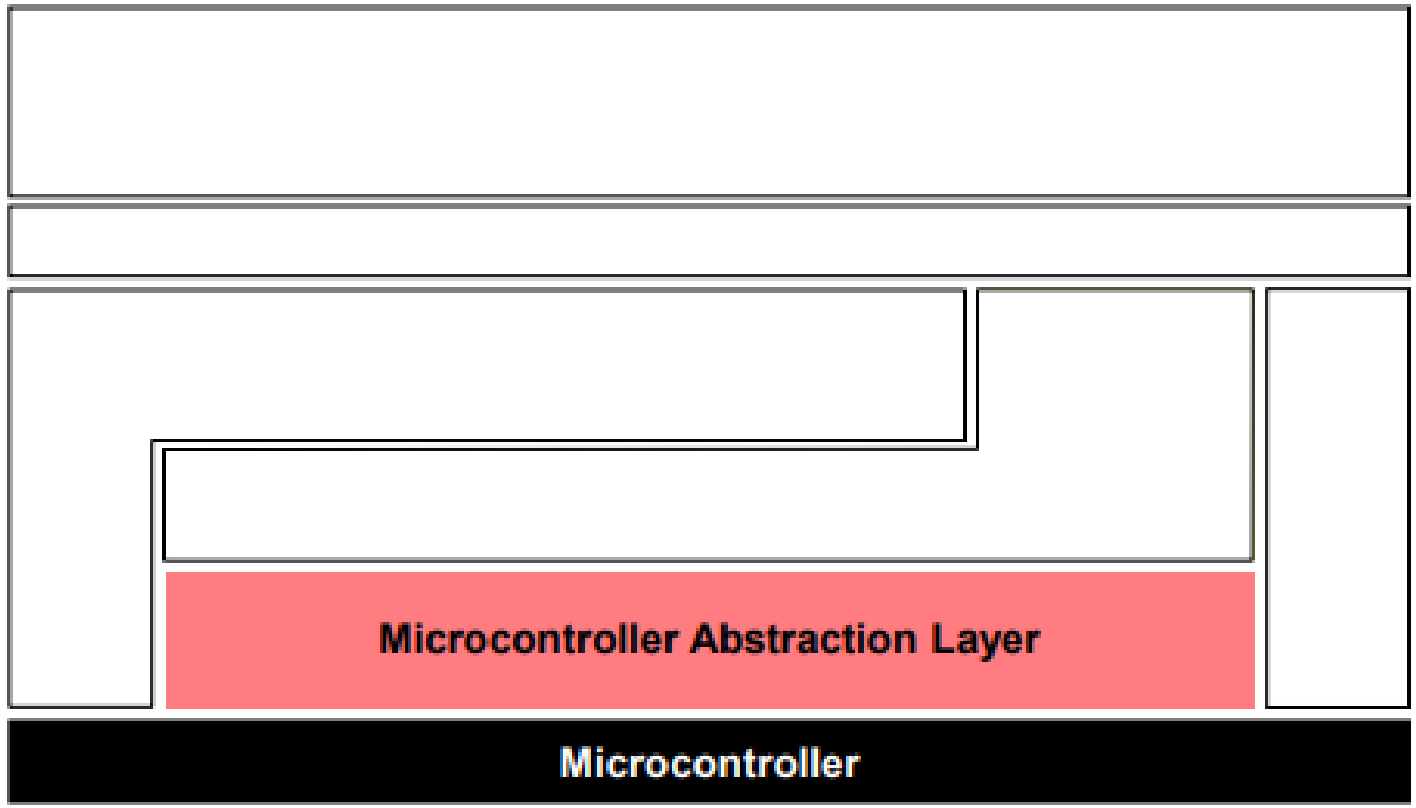
AUTOSAR BSW

- The AUTOSAR Basic Software is further divided in the layers: Services, ECU Abstraction, Microcontroller Abstraction and Complex Drivers.



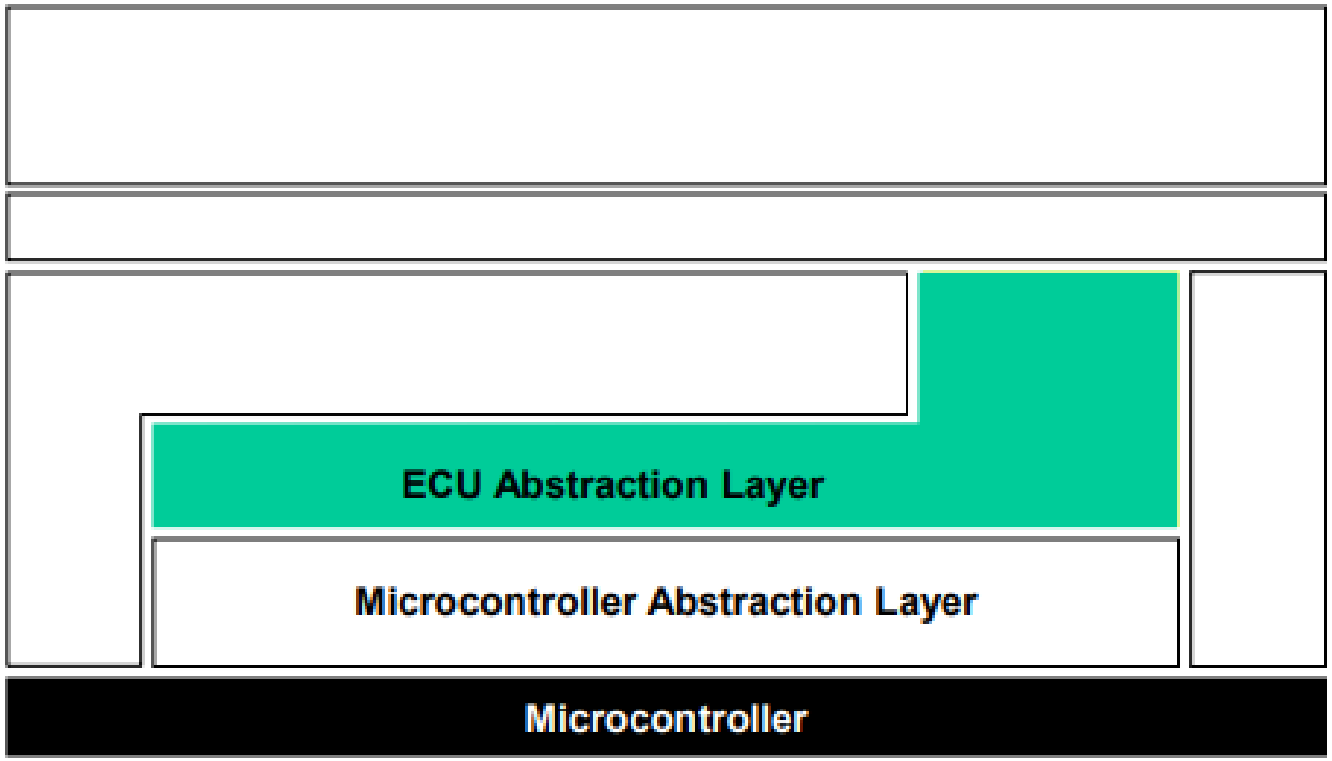
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- The Microcontroller Abstraction Layer is the lowest software layer of the Basic Software. It contains internal drivers, which are software modules with direct access to the µC and internal peripherals.



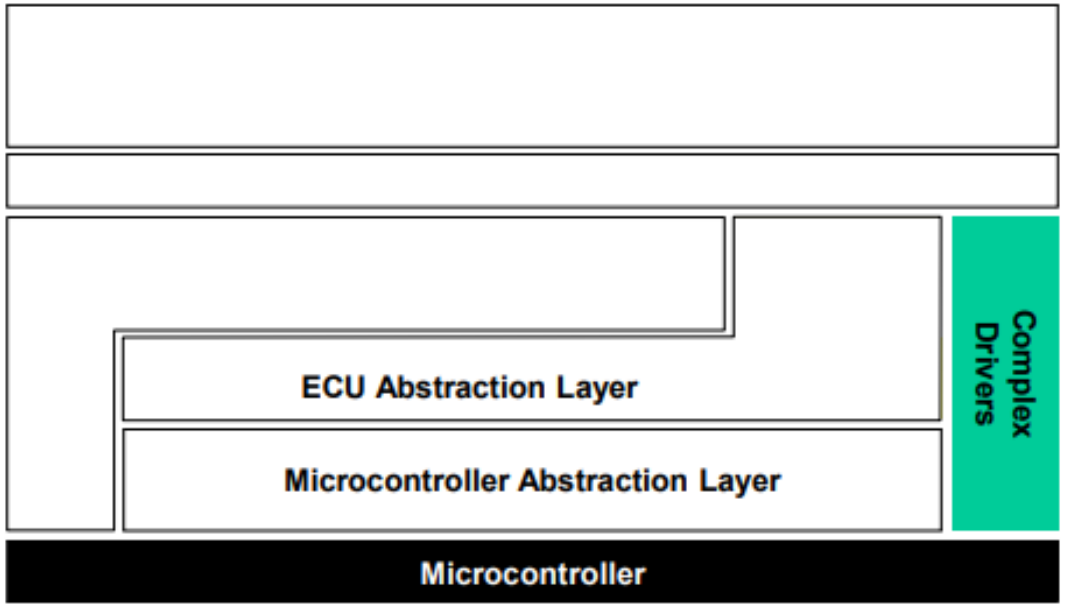
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- The ECU Abstraction Layer interfaces the drivers of the Microcontroller Abstraction Layer. It also contains drivers for external devices. It offers an API for access to peripherals and devices regardless of their location (µC internal/external) and their connection to the µC (port pins, type of interface)



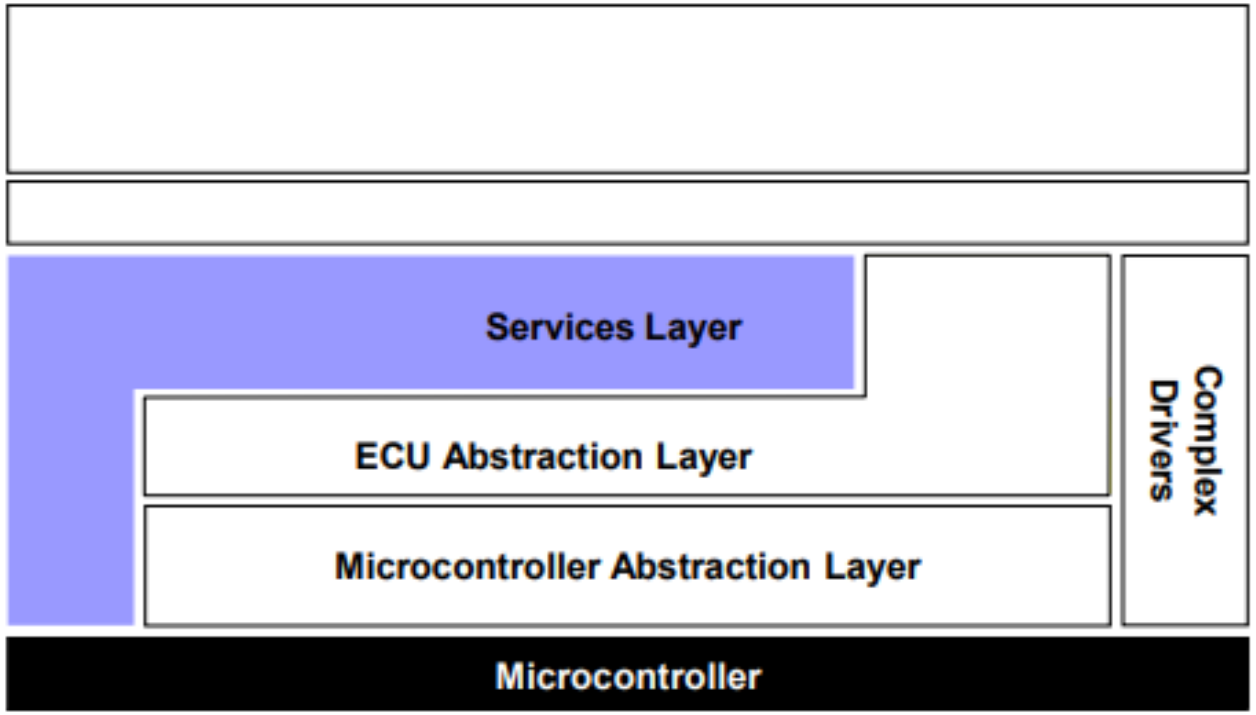
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- The Complex Drivers Layer spans from the hardware to the RTE.
- Provide the possibility to integrate special purpose functionality, e.g. drivers for devices:
  - which are not specified within AUTOSAR,
  - with very high timing constrains or
  - for migration purposes etc.



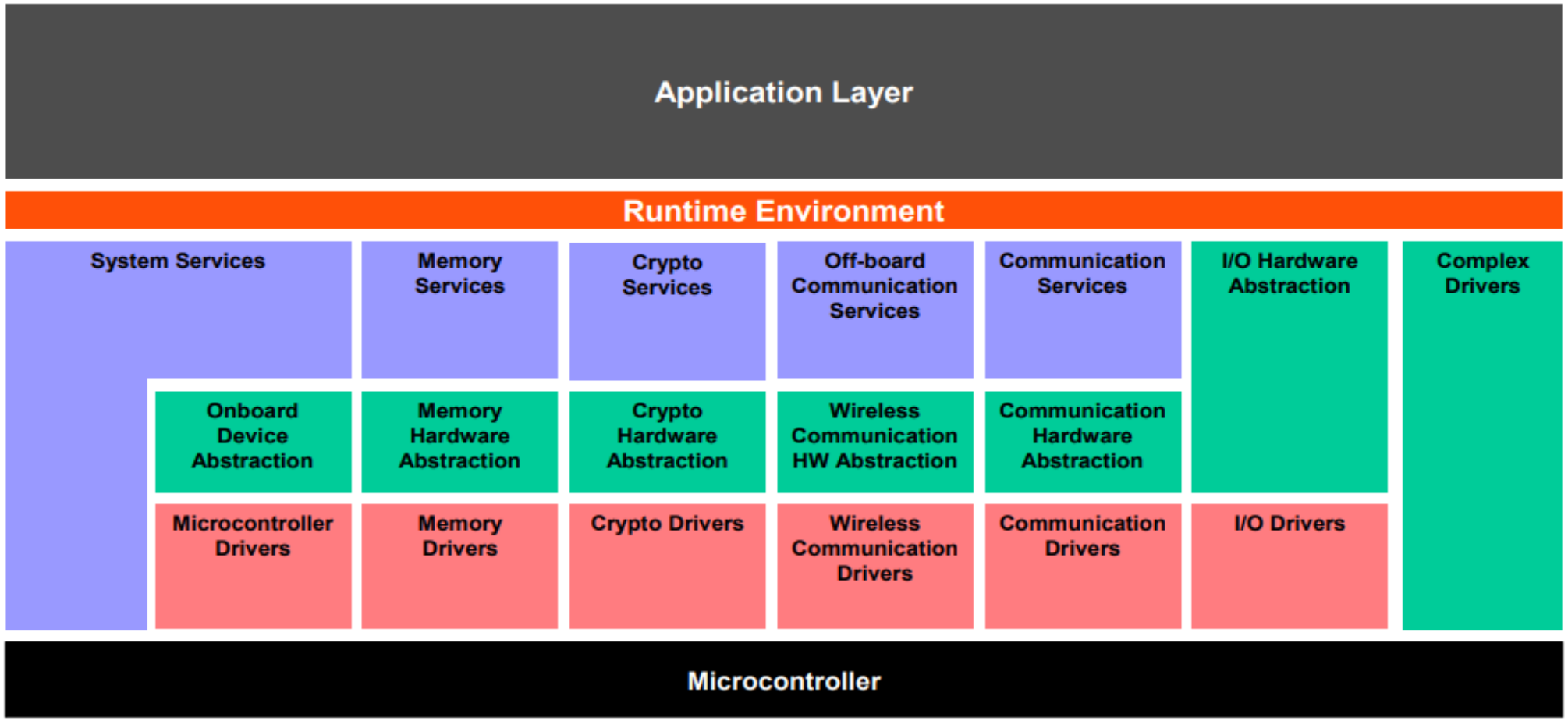
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- The Services Layer is the highest layer of the Basic Software which also applies for its relevance for the application software: while access to I/O signals is covered by the ECU Abstraction Layer.



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- The Basic Software Layers are further divided into functional groups. Examples of Services are System, Memory and Communication Services.



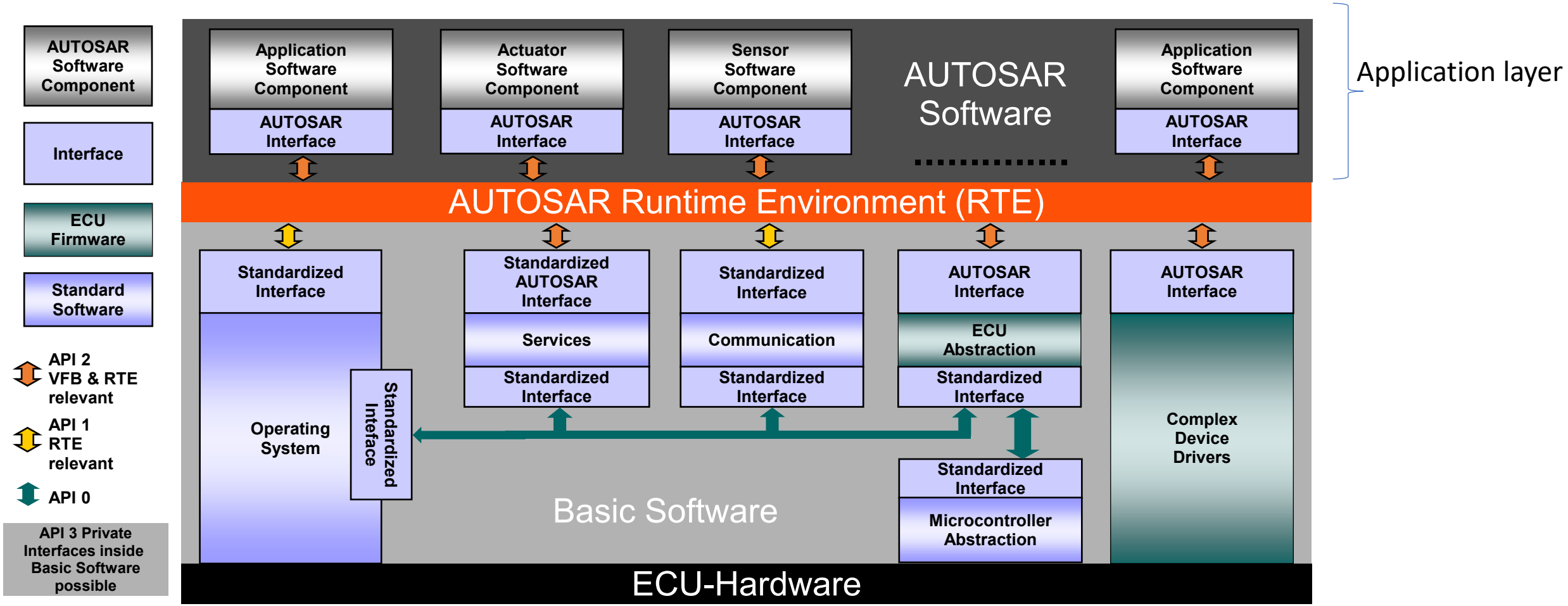
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- The Basic Software can be subdivided into the following types of services:
  - **Input/Output (I/O):** Standardized access to sensors, actuators and ECU onboard peripherals
  - **Memory:** Standardized access to internal/external memory (non volatile memory)
  - **Crypto:** Standardized access to cryptographic primitives including internal/external hardware accelerators
  - **Communication:** Standardized access to: vehicle network systems, ECU onboard communication systems and ECU internal SW
  - **Off-board Communication:** Standardized access to: Vehicle-to-X communication, in vehicle wireless network systems, ECU off-board communication systems
  - **System:** Provision of standardizeable (operating system, timers, error memory) and ECU specific (ECU state management, watchdog manager) services and library functions

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AUTOSAR CLASSIC PLATFORM

LAYERED ARCHITECTURE



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AUTOSAR ADAPTIVE PLATFORM

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AUTOSAR ADAPTIVE PLATFORM

WHY DO WE NEED A NEW STANDARD

Mobility



Electrification



Automated driving



New functionalities such mentioned above, have introduced a new set of problems to the electrical/electronic (E/E) panorama. These new functionalities differ from the ones defined and supported by Classic AUTOSAR

Specific infrastructure and SW/HW architecture need to be defined, that’s why they introduce new standard which is **Adaptive AUTOSAR**

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AUTOSAR ADAPTIVE PLATFORM

WHY DO WE NEED A NEW STANDARD

- The AUTOSAR Classic Platform (CP) standard addresses the needs of deeply embedded ECUs, while the needs of ECUs described in the previous slide cannot be fulfilled well. Therefore, AUTOSAR specifies a second software platform, the **AUTOSAR Adaptive Platform (AP).**

- AP provides mainly high-performance computing and communication mechanisms and offers flexible software configuration, e.g., to support software update over-the-air.

- Features specifically defined for the CP, such as access to electrical signals and automotive specific bus systems, can be integrated into the AP, but is not in the focus of standardization.

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AUTOSAR ADAPTIVE PLATFORM

AUTOSAR CLASSIC VS ADAPTIVE

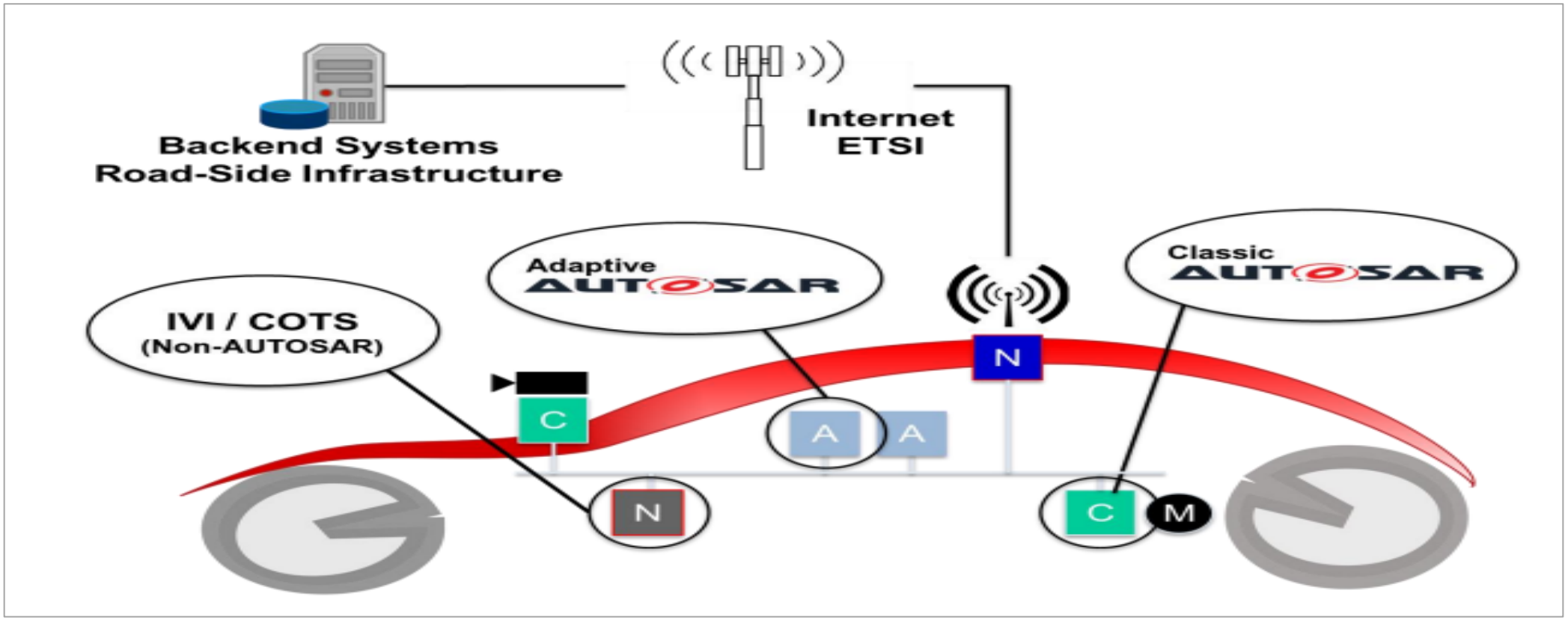
Goal or Feature	Classic Platform	Adaptive Platform
Use cases	Embedded systems	High performance computing, communication with external resources, and flexible deployment
Programming language	C	C++
Operating system	Bareboard	POSIX
Real-time requirements	Hard	Soft
Computing power	Low	High
Communication	Signal-based	Event-based, service-oriented
Safety and security	Supported	Supported
Dynamic updating	Not available	Incremental deployment and run-time configuration changes
Level of standardization	High—detailed specifications	Low—APIs and semantics
Agile development	No	Yes

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AUTOSAR ADAPTIVE PLATFORM

INTEGRATION OF CLASSIC, ADAPTIVE AND NON-AUTOSAR ECUS

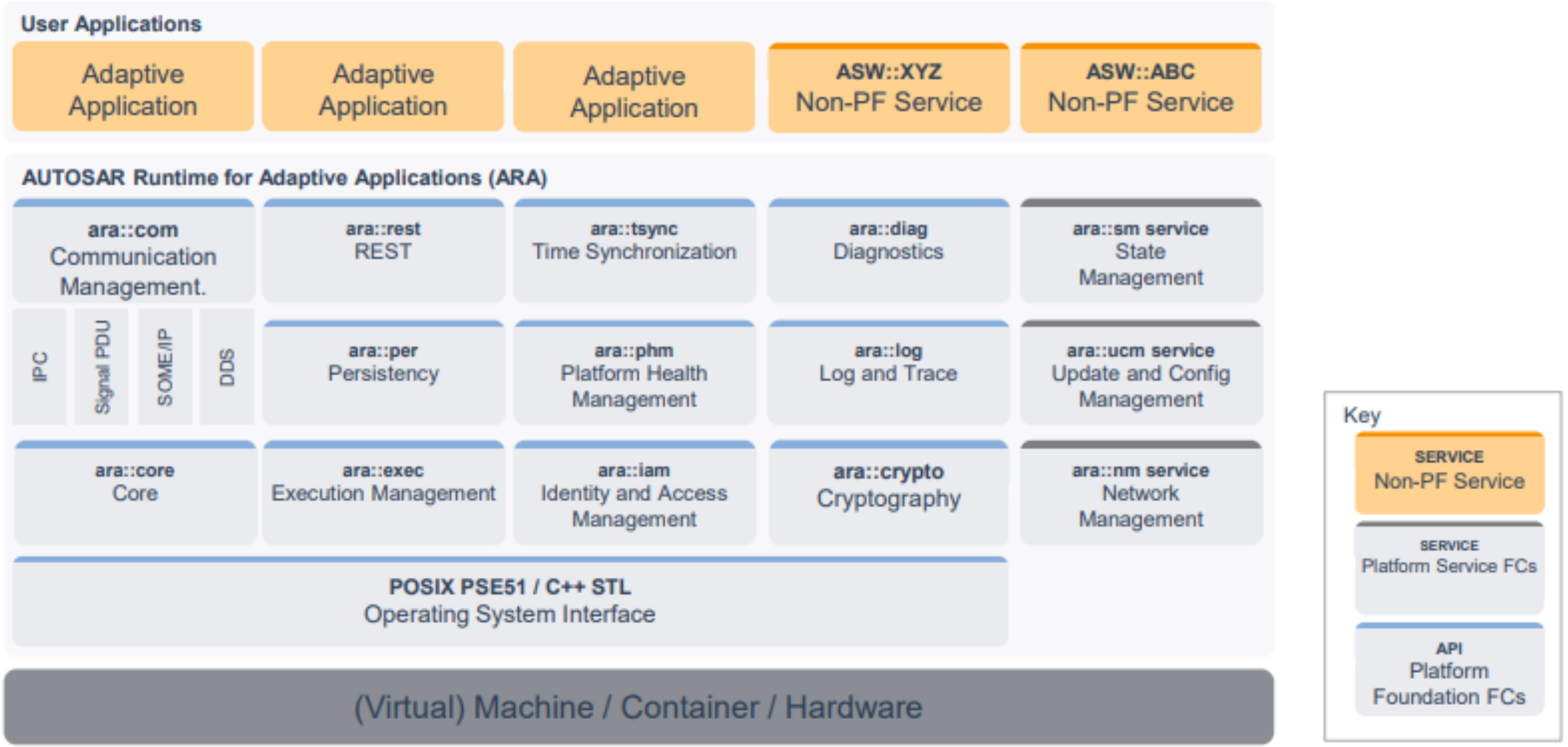
AP will not replace CP or Non-AUTOSAR platforms in IVI/COTS. Rather, it will interact with these platforms and external backend systems such as road-side infrastructures, to form an integrated system



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AUTOSAR ADAPTIVE PLATFORM

ARCHITECTURE



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AUTOSAR ADAPTIVE PLATFORM

ARCHITECTURE

- The Adaptive Applications (AA) run on top of ARA, AUTOSAR Runtime for Adaptive applications.
- ARA consists of application interfaces provided by Functional Clusters, which belong to either Adaptive Platform Foundation or Adaptive Platform Services.
- Adaptive Platform Foundation provides fundamental functionalities of AP, and platform standard services called Adaptive Platform Services.
- Any AA can also provide Services to other AA, illustrated as Non-platform service in the figure.

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AUTOSAR ADAPTIVE PLATFORM

FUNCTIONAL CLUSTER CATEGORIES

Runtime

- Execution management
- State management
- Log and Trace
- Core
- Operating system interface

Communication

- Communication management
- Network management
- Time Synchronization

Storage

- Persistency

Security

- Cryptography
- Identity and access management
- Adaptive instruction Detection system manager

Safety

- Platform Health management

Configuration

- Update and Configuration management
- Registry

Diagnostics

- Diagnostic management

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AUTOSAR ADAPTIVE PLATFORM

RUNTIME FUNCTIONAL CLUSTER

1. Execution management

- Execution Management is responsible to control Processes of the AUTOSAR Adaptive Platform and Adaptive applications. It starts, configures, and stops Processes as configured in Function Group States using interfaces of the Operating System.
- Execution Management then controls the startup and shutdown of the AUTOSAR Adaptive Platform

2. State management

- State Management determines the desired target state of the AUTOSAR Runtime for Adaptive Applications based on various application-specific inputs.

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3. Log and Trace

- Log and Trace provides functionality to build and log messages of different severity.
- An Adaptive Application can be configured to forward log messages to various sinks, for example to a network, a serial bus, the console, and to non-volatile storage.

4. Core

- Core provides functionality for initialization and de-initialization of the AUTOSAR Runtime for Adaptive Applications as well as termination of Processes.

5. Operating system interface

- The Operating System Interface provides functionality for implementing multi-threaded real-time embedded applications and corresponds to the POSIX PSE51 specification.
- Asynchronous (real-time) signals and message passing enable inter-process communication.
- High resolution timers and clocks are provided to control real-time behavior precisely.

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1. Communication management

Communication Management is responsible for all levels of service-oriented and raw communication between applications in a distributed real-time embedded environment.

2. Network management

Network Management provides functionality to request and query the network states for logical network handles, Such network handles can be mapped to physical or partial networks.

3. Time synchronization

- Time Synchronization provides synchronized time information in distributed applications.
- Synchronized time information between different applications and/or Machines is of paramount importance when the correlation of different events across a distributed system is needed, either to be able to track such events in time or to trigger them at an accurate point in time.

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Persistency

Persistency offers mechanisms to applications to store information in the non-volatile memory of an Adaptive Platform Instance. The data is available over boot and ignition cycles. Persistency offers a library-based approach to access the non-volatile memory.

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1. Cryptography

Cryptography provides various cryptographic routines to ensure confidentiality of data, to ensure integrity of data (e.g., using hashes), and auxiliary functions for example key management and random number generation.

2. Identity and access management

- Identity and Access Management checks access to resources of the AUTOSAR Adaptive Platform, for example, on Service Interfaces and Functional Clusters.
- Identity and Access Management hereby introduces access control for Adaptive Applications and protection against privilege escalation in case of attacks.

3. Adaptive Intrusion Detection System Manager

Adaptive Intrusion Detection System Manager provides functionality to report security events.

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Platform Health Management

Platform Health Management performs supervision of Processes in safety-critical setups and reports failures to State Management.

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Update and Configuration Management

- Update and Configuration Management is responsible for updating, installing, removing and keeping a record of the software on an AUTOSAR Adaptive Platform in a safe and secure way.
- It enables to update the software and its configuration flexibly through over-the-air updates (OTA).

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Diagnostic Management

- Diagnostic Management is responsible for handling diagnostic events produced by the individual Processes running in an AUTOSAR Runtime for Adaptive Applications. Diagnostic Management stores such events and the associated data persistently according to rendition policies.
- Diagnostic Management also provides access to diagnostic data for external Diagnostic clients via standardized network protocols (ISO 14229-5 (UDSonIP) which is based on the ISO 14229-1 (UDS) and ISO 13400-2 (DoIP)).

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QUESTIONS

primatec engineering

QUESTIONS

1. What are the limitation and challenges of legacy model of SW/HW developement ?
2. What's AUTOSAR ?
3. What are the main advantages of AUTOSAR?
4. What are the different E/E architectures ?
5. Present the layered architecture of Classic AUTOSAR and describe briefly different layers
6. What's a software component (SWC) and its main characteristics ?
7. What's a the role of RTE ? and present deeply the different types of communication managed through RTE
8. Explain what's BSW in classic AUTOSAR
9. What's the role of service layer in CP and provide some examples
10. How to choose the type of communications (Client/Server or Sender receiver) ? Explain more by providing some examples ?

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QUESTIONS

11. What’s VFB ?
12. What’s Complex drivers in BSW layer and why they introduced it ?
13. What’s the difference between Classic and adaptive AUTOSAR ?
14. Why Adaptive AUTOSAR was introduced ?
15. Present and explain at least 4 Functional clusters in adaptive AUTOSAR

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